UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the May/June 2007 question paper

5070 CHEMISTRY

5070/04 Paper 4 (Alternative to Practical), maximum raw mark 60

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1 **(a)** syringe [1]

(b) 72 cm^3 [1]
[Total: 2]

2 **(a)** lime water turns milky [1]

(b) **(i)** effervescence stopped [1]

(ii) solid remained
(answers must be different in **(i)** and **(ii)**)
(no precipitates or solubility in answers) [1]

(c) filtration [1]

(d) (iii) [1]

(e) 0.025 moles [1]

(f) 161 × 0.025 = 4.025 (1) g [1]

(g) 0.6 dm^3 [1]
[Total: 8]

3 **(a)** **(i)** bromine, brown gas, colour spreads to fill all the jar [1]

(ii) diffusion [1]

(b) **(i)** brown to colourless [1]

(ii) it contains a double bond or is unsaturated [1]

(iii) addition [1]

(iv) C_3H_6 (1), propene (1)
(no consequential name mark for incorrect alkene) [2]
[Total: 7]

4 **(a)** [1]
[Total: 1]

5 **(c)** [1]
[Total: 1]

6 **(c)** [1]
[Total: 1]

7 **(d)** [1]
[Total: 1]

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8 **(a)** 1.51 (g) [1]

(b) pipette [1]

(c) purple or pink to colourless [1]

(d)

25.2	31.1	48.3	mark correct rows OR columns to advantage of candidate. One mark for each correct row OR column (3)
0.0	6.8	23.8	
25.2	24.3	24.5	

mean titre = 24.4 cm^3 (4) [4]

(e) 0.00244 (moles) [1]

(f) 0.00244 [1]

(g) 0.0244 [1]

(h) 0.05 [1]

(i) 0.0256 [1]

(j) 0.0128 [1]

(k) 118 [1]

(l) x = 2 (1), = 4 (1) [2]

(m) **(i)** $C_2H_5\ OOC\ C_2H_4\ COOC_2H_5$ [1]

(ii) esters [1]

[Total: 18]

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9 **(a)** transition metals or metal ions not present
(Not '**V** is not a transition metal') [1]

(b) **(i)** white ppt., [1]

(ii) soluble in excess [1]

(c) **(i)** aqueous ammonia [1]

(ii) white ppt. [1]
excess aqueous ammonia, soluble in excess [1]
(use of ammonia or ammonium loses first mark)

(d) dilute nitric acid (1), aq. silver nitrate (1) white ppt. (1) [3]
[Total: 9]

10 **(a)** temperatures – 27.5, 29.9, 32.4, 33.0 all correct [1]
temperature rises – 2.5, 4.9, 7.4, 8.0 all correct [1]

(b) all points plotted correctly (1)
two straight lines drawn in <u>each</u> case intersecting correctly (2)
(points joined by a curve or third straight line in each case, 1 only)
through zero (1) [4]

(c) 0.32 g [1]

(d) 0.56 g [1]

(e) **(i)** 0.65 g [1]

(ii) 0.70 g [1]

(f) No temperature change or no reaction (1), Ag below Cu in reactivity series or silver does not react with copper(II) sulphate (1)
(any suggestion of a reaction or temperature change loses both marks) [2]

The answers to **(c)**, **(d)** and **(e)** are those on the candidate's graph.
In all cases read to nearest HALF SMALL SQUARE. **[Total: 12]**

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